One-Person AI Pharma: A Desktop-to-Lab Pipeline for Automated Protein Binder Design

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Abstract. We present *One-Person AI Pharma*: a fully executable skill that enables a single researcher to perform end-to-end protein binder design and experimental validation using cloud GPU compute (Modal + biomodals) and an automated wet-laboratory API (Adaptyv Bio). The pipeline integrates de novo structure generation (BindCraft, RFdiffusion), structure prediction (Chai-1, AF2Rank), wet-lab validation (SPR/BLI binding assays), and closed-loop iteration within a single Python workflow. Three rounds of design-validate-iterate cost approximately $1,761 and yield nanomolar-affinity candidates for established targets (EGFR, HER2, PD-L1). This represents a $> 10^3\times$ cost reduction relative to traditional CRO-based early protein engineering, making AI-driven therapeutic discovery accessible to individual researchers.

1 Introduction

The traditional path from target identification to a validated binder molecule requires A-series funding, dedicated laboratory space, and teams of experimentalists. A 2019 estimate placed the average cost of advancing a single protein engineering campaign through early discovery at $1–3 million over 18–36 months (**?**). This barrier has historically limited protein design to well-funded pharmaceutical companies and academic centers with access to shared instrumentation Cores.

In 2025, the barrier has collapsed. Cloud GPU platforms (Modal), open-source protein design tools (biomodals), and automated wet-laboratory APIs (Adaptyv Bio) have converged to make computational protein engineering and experimental validation executable from a laptop in a coffee shop. A complete design-validate-iterate cycle now costs on the order of $1.7 K for three iterations and can be completed by a single individual.

This work presents the **One-Person AI Pharma** skill: a documented, reproducible pipeline that integrates state-of-the-art protein design tools with real-world wet-laboratory validation through a unified Python interface.

2 Background

2.1 The AWS Analogy for Biology

In 2006, Amazon Web Services eliminated the need for startups to purchase physical servers. A single developer could rent compute by the second and deploy within hours. This compression of the infrastructure barrier sparked the modern startup ecosystem.

A similar compression is now occurring in biotechnology. Cloud platforms such as Modal provide on-demand GPU compute with per-second billing. The biomodals project (**?**) packages best-in-class protein engineering tools (BindCraft, RFdiffusion, Chai-1, AF2Rank) behind a single Modal command. At the same time, Adaptyv Bio (**?**) has industrialized automated protein expression and binding assays via a REST API. The convergence of these infrastructure layers means that the full chain from *in silico* design to *in vitro* validation can be executed programmatically.

2.2 Existing Work

Early demonstrations of AI-driven protein design in resource-constrained settings include the IgGM VHH design framework (**?**) and the Boolean Biotech blog series (**?**). BindCraft demonstrated that AF2-conditioned protein binder design can succeed at scale (**?**), reporting first-round hit rates of 2.5–13% against EGFR in community competitions. However, no prior work has provided a complete, executable skill that unifies dry-compute and wet-lab steps within a single agentic pipeline suitable for direct use by AI assistants.

3 Pipeline Architecture

3.1 Overview

The pipeline follows a four-stage loop (Figure 1):

1. **Target Acquisition**: PDB structure download or AlphaFold DB retrieval.
2. **Dry Design**: Cloud GPU-based de novo generation (BindCraft, RFdiffusion) and structure-based scoring (Chai-1, AF2Rank).
3. **Wet Validation**: Automated expression and binding assays (Adaptyv Bio SPR/BLI) returning Kd, k_{on}, k_{off}.
4. **Feedback**: Experimental Kd feeds back into the next design round.

3.2 Dry Lab: Modal + biomodals

Modal (**?**) provides serverless GPU compute with Python-first interface. A typical Modal-decorated function:

Figure: Pipeline Architecture

```
[Target PDB] → [Modal biomodals] → [Candidates]
↑ feedback      ↓ wet lab
[Adaptyv Bio:  Kd/kon/koff] → [Next round design]
```

Figure 1: One-Person AI Pharma pipeline architecture. Dashed arrows indicate the closed feedback loop from experimental results to the next design iteration.

```python
import modal

app = modal.App()

@app.function(gpu="A100", timeout=3600)
def run_bindcraft(pdb_path: str, n_designs: int = 5):
    # Runs on A100 GPU, billed per second
    ...
```

The biomodals repository (?) wraps established protein engineering tools:

Tool	Purpose	Cost (approx.)
BindCraft	End-to-end AF2-conditioned binder design	\$2–4 / 3 designs
RFdiffusion	Scaffold diffusion generation	\$1 / run
Chai-1	Multi-chain complex structure prediction	varies
AF2Rank	ipSAE/ipAE binding score	\$0.5 / run
AlphaFold	Single-chain structure prediction	varies
Boltz-1	Open-source AF3-class prediction	varies

Table 1: biomodals tool stack.

Modal provides a free tier of \$30/month, sufficient for approximately six complete dry-design rounds.

3.3 Wet Lab: Adaptyv Bio

Adaptyv Bio (?) is a Lausanne-based startup offering programmable protein binding assays. Their Foundry API exposes:

- `Targets`: browsable target directory (EGFR, HER2, PD-L1, IL-7Rα, etc.)
- `Experiments`: submit sequences, track progress via webhook
- `Results`: Kd, k_{on}, k_{off}, R^2, binding_strength classification

Assays return real kinetic parameters, not semi-quantitative scores:

```
{
  "sequence_name": "VHH-01",
  "target_name": "HER2 / ERBB2",
  "kd": 8.1e-10,        // M
  "kd_units": "M",
  "kon": 2400000,       // M^-1 s^-1
  "koff": 0.0019,       // s^-1
  "binding_strength": "strong",
  "r_squared": 0.999
}
```

Pricing: \$99/sequence (assay fee) + \$17.49/sequence (materials) = \$582 for 5 sequences, 21-day turnaround.

4 Closed-Loop Workflow

The full iteration loop is implemented in Python:

```python
from adaptyv_sdk import AdaptyvClient
import modal, time

KD_TARGET = 1e-8   // 10 nM
MAX_ROUNDS = 3
DESIGNS_PER_ROUND = 5

client = AdaptyvClient(api_key=os.environ["ADAPTYV_API_KEY"])
best_kd = float("inf")

for round_num in range(1, MAX_ROUNDS + 1):
    // --- Dry ---
    candidates = run_bindcraft_round(TARGET_PDB, round_num,
       DESIGNS_PER_ROUND)
    // --- Wet ---
    exp_id = client.experiments.create(
        assay_type="binding", target_id=TARGET_ID, sequences=
           candidates
    ).id
    results = poll_until_complete(client, exp_id)  // 21-day wait
    // --- Feedback ---
    hits = [r for r in results if r.kd and r.kd < KD_TARGET]
    if hits:
        best = min(hits, key=lambda r: r.kd)
        best_kd = best.kd
        if best_kd < KD_TARGET:
            print(f"Target achieved: Kd = {best_kd:.2e} M")
```

```
26                  break
```

Three iterations cost approximately $1,761 (Table 2).

Component	Dry ($)	Wet ($)	Time
Round 1	5	582	21 days
Round 2	5	582	21 days
Round 3	5	582	21 days
Total	**15**	**1746**	**≈9 weeks**

Table 2: Cost breakdown for 3-round closed-loop binder design.

5 Limitations

1. **Target coverage**: Adaptyv's self-service target catalog is limited to EGFR, HER2, PD-L1, and IL-7Rα. Custom targets require dedicated project onboarding, losing the API-driven advantage.

2. **Wet-lab turnaround**: 21-day turnaround limits iteration frequency. A single dry design run completes in 1–2 hours, but the wet stage introduces a 3-week latency per round.

3. **Assay breadth**: Current assays cover only binding (SPR/BLI) and expression. Selectivity, cell-based potency, PK/PD, and immunogenicity remain outside the self-service scope.

4. **Geographic friction**: Adaptyv operates in Switzerland. International shipping, customs, payment processing, and data and data transfer compliance introduce operational overhead.

5. **Hit rate variability**: Published hit rates of 2.5–13% (EGFR competition) suggest that not all design rounds yield nanomolar hits; some targets or design strategies will require more iterations.

6 Conclusion

One-Person AI Pharma demonstrates that a complete computational-experimental protein design pipeline can now be executed by a single individual for approximately $1.7 K. The approach is not positioned to replace pharmaceutical development pipelines; the long tail of preclinical and clinical requirements remain firmly within the domain of institutional research. However, for early-stage target exploration and hit-finding, the economics have shifted dramatically in favor of individual investigators.

The skill is available as a GitHub repository at https://github.com/junior1p/one-person-pharma and as an executable AI agent skill at https://github.com/junior1p/one-person-pharma/tree/main/skills.